

Robert Tylman

EXPERIENCE

Camp Solelim, Sudbury, Canada — Assistant Director, Full-Stack Developer

May 2021 – August 2025

- Developed a professional “canoe trip planning system” leveraging Google Earth API for route visualization, campsite selection, and meal planning, saving \$10,000+ per summer and increasing canoe trip enrollment by 550%.
- Designed and maintained *CampSorter*, a full-stack scheduling and logistics platform (JavaScript, Python, SQL), reducing admin workload by 70% and improving camp programming efficiency; system remains in active use.

People’s Trust Company, Toronto, Canada — Software Engineer & IT Analyst Intern

May 2022 – August 2022

- Automated legacy payroll workflows using Python pipelines, eliminating manual processing for high-volume credit datasets and reducing errors by 95%.
- Managed and secured company-wide IT infrastructure, including user permissions, networking, encryption, and security protocols, protecting sensitive financial and employee data.

MyPart, Tel Aviv, Israel — Machine Learning Intern

September 2020 – January 2022

- Designed full-feature Music Information Retrieval (MIR) models for critical audio tasks - including stem separation and harmonic/rhythm/tempo extraction - for the analysis of artist catalogues.
- Trained and labelled large-scale existing MIR models on 50,000+ audio tracks for pattern recognition and automated metadata classification, increasing the relevance of content-based search results by 15% across music libraries.

PROJECTS

Twist!  — *Expo, Express, Gemini 3.0, Node.js, React.js, Redis, Socket.io*

- Released an original, full-stack, multiplayer party game that uses the Nano-Banana Pro to transform player camera-roll images into AI-powered challenges, acquiring 500+ active users within the first month.
- Engineered a real-time image-processing and voting pipeline delivering sub-100ms latency across all game mechanics, ensuring a seamless, responsive UI on both web and mobile devices (Expo/React.js).
- Designed a robust and scalable backend architecture using Redis for state management and Socket.io for real-time communication, supporting all concurrent users with high reliability and low latency.

Chrome Recorder  — *TypeScript, React, Web Audio API, IndexedDB*

- Released a high-performance Chrome Extension that allows you to record and edit audio in your browser.
- Built a real-time waveform visualizer, lossless audio pipeline, and multi-file-format exporting system, enabling smooth playback and customization for storage.

Audio Plugin Suite — *C++, JUCE*

- Developed a suite of DAW plugins (Synths, EQs, Filters) for professional audio production, implementing advanced signal processing algorithms with intuitive, production-ready interfaces.
- Optimized for low-CPU and low-latency, enabling seamless integration into professional production workflows.

Sony WH-1000XM6 Headphones Collaboration

- Partnered with Sony AI and engineering teams to analyze and optimize ML-driven noise cancellation algorithms for flagship headphones.
- Provided detailed technical assessments of ANC algorithm performance, acoustic behavior, and improvement opportunities, contributing to research-driven product innovation.

EDUCATION

New York University, New York City, NY

2025 – Present

- Master’s in Machine Learning: DSP, MIR, Music Technology
- Focus: Applied ML for audio signal processing
- Research: Integrating LLMs with music software for enhanced automation and interaction.

Vanderbilt University, Nashville, TN

2021 – 2025

- BS in Computer Science (Engineering), Concentration in DSP
- GPA: 3.83/4.00, (*Cum Laude, Dean’s List*)
- Winner of the 2023 *Blair School Of Music Award For Innovative Creative Work in Audio Technology*

SKILLS

- **Programming Languages:** C++, Expo, HTML, Java, JavaScript, Node.js, Python, R, React, Rust, Socket.io, TypeScript
- **Tools & Technologies:** DAWs, Docker, Git, Keras, Librosa, Max/MSP/Jitter, Pro Tools, PyTorch, SQL, Tensorflow